YEAMIN MAHMUD

Research Engineer

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LinkedIn | GitHub



Professional Summary

CS graduate with experience in C++ development, codebase analysis, and software integration. Hands-on experience with Arduino-based embedded systems, sensor integration, and real-time data collection through academic and research projects. Skilled in C/C++, Python, debugging, and object-oriented programming, with a strong foundation in data structures, algorithms, and analytical problem-solving. Interested in developing reliable embedded software and hardware-integrated solutions.

Technical Skills

Programming Languages	C, C++, Python, Java, PHP
Embedded Systems	Arduino Uno, Microcontroller-Based Development, Hardware-Software Integration
Sensors and Hardware	MAX30102, LSM9DS1, Sensor Integration, Real-Time Data Collection
Programming Concepts	Object-Oriented Programming, Data Structures, Algorithms, Debugging, Codebase Analysis
Software Development	Software Implementation, Troubleshooting, Technical Documentation, API Integration
Databases	PostgreSQL, MySQL
Development Tools	Git, GitHub, Docker, Linux, Visual Studio Code, Postman
Additional Technologies	FastAPI, Django, TensorFlow, PyTorch

Professional Experience

TOI DLT Foundation *JUN 2025 – Present*
Research Engineer Bangladesh

- Develop and maintain core gateway modules and Remote Procedure Call (RPC) functions for blockchain-based infrastructure.
- Analyze existing codebases, investigate software issues, and implement improvements to support system reliability and maintainability.
- Contribute to the development and integration of SMC Protocol components and research related to ACAN.
- Prepare technical documentation and developer guides covering implementation details and protocol integration.
- Collaborate on technical tasks and troubleshoot implementation issues within existing software systems.

ATI Limited Bangladesh *JAN 2025 – APR 2025*
Python Developer Intern Bangladesh

- Designed and documented FastAPI endpoints for enterprise integration projects.
- Participated in Odoo 18 implementation and business workflow customization through Project Leader Onboarding training.
- Contributed to application development and publishing an application on the Odoo Official Marketplace.

North South University, DIAL Lab *SEP 2022 – JAN 2023*
Research Assistant Dhaka, Bangladesh

- Collaborated on wearable healthcare technology research involving sensor-based hardware systems.
- Performed data collection, analysis, and technical documentation for sensor-based research projects.
- Contributed to research methodologies and academic publication activities.

Education

BRAC University *2025 – Present*
M.Sc. in Computer Science and Engineering Dhaka, Bangladesh

North South University *April 2025*
B.Sc. in Computer Science and Engineering Dhaka, Bangladesh

- Academic result: 82%.
- Relevant coursework: Data Structures and Algorithms, Object-Oriented Programming, Software Engineering, Operating Systems, Artificial Intelligence, and Machine Learning.

Selected Technical Projects

Multi-Sensor Wearable Health Watch GitHub
Technologies: C++, Arduino Uno, GPS, MAX30102, LSM9DS1

- Integrated multiple hardware sensors, including MAX30102 and LSM9DS1, into a wearable monitoring system using Arduino Uno.
- Implemented real-time data collection and processing for health and motion monitoring.
- Worked on hardware-software integration and sensor data handling for a wearable healthcare prototype.

Mars Terrain Analysis and Classification GitHub

Technologies: Python, TensorFlow Quantum, Cirq, Jupyter Notebook

- Developed quantum neural network models for Martian surface classification using machine learning techniques.
- Compared classical and quantum approaches to investigate model performance and alternative methods.

Local Connect – Healthcare Locum Marketplace GitHub

Technologies: React, Vite, TypeScript, Supabase, Stripe

- Contributed to a healthcare marketplace connecting locum clinicians with practices through a web application.
- Collaborated with a co-founder on Stripe payment gateway integration and implementation of marketplace payment workflows.
- Developed a Supabase Edge Function for batch processing of professional verification records.

Poker Simulator – Game Theory Application GitHub

Technologies: Python, NumPy, Pandas, Matplotlib

- Implemented an Expectiminimax algorithm for decision-making under probabilistic conditions.
- Executed simulations and analyzed strategic outcomes using data processing and visualization.

Achievements

- Innovations Challenge NSU 2025: Ranked among the top 15 teams out of 150+ participating teams.
- NHSPC Regional Championship: Secured first place in a programming competition.

Certifications

- Machine Learning with Python – Coursera (2024).
- Quantum Computing Foundations – IBM Quantum (2024).
- Linear Algebra – MIT OpenCourseWare (2023).
- Odoo 18 Project Leader Onboarding Training – ATI Limited (2025).

References

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